AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2019

Computer Studies Paper II

Time: 2 hours 20 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

RUBRIC

2. There are SEVEN questions. Answer ALL questions. Question 7 offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. Aslam downloaded photo albums from his social media account. If the average size of each photo is 512 kilobyte, then how many photos can be stored in 75 megabyte storage space? (2 Marks)

b. Differentiate between media and devices. Give any ONE example of storage media. (2 Marks)

Q.2. (Total 5 Marks)

a. Convert $(A9B)_{16}$ into decimal number system. (3 Marks)

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b. Define the following two types of programming languages.

i. Low level programming language (1 Mark)

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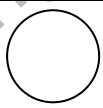
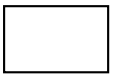
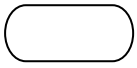
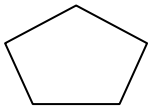
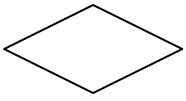
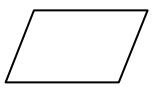
ii. High level programming language (1 Mark)

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Q.3. (Total 5 Marks)

A program flowchart consists of different symbols and each symbol has a specific purpose. A list of symbols and purposes is given in the following table. Use an arrow (→) to connect each purpose with relevant symbol.

Purpose	Symbol
Processing	
Input/ Output	
Connector	
Decision making and branching	
Start/ End of a Flowchart	
	

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Q.4. (Total 4 Marks)

Write a GW-BASIC program that will evaluate the given polynomial for any input value of *x* and show the result.

$5x^3 - 3x^2 + 9$

Q.5. (Total 4 Marks)

a. Differentiate between the following: (2 Marks)

Conditional transfer of control	Unconditional transfer of control

IF…THEN…ELSE statements	FOR…NEXT statements

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b. If the subscript of the first element of an array **X** is (0, 0), then write GW-BASIC syntax for a two dimensional array to store the given sequence of numbers. (2 Marks)

11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121, 132

Q.6. (Total 5 Marks)

Write a GW-BASIC program that

- inputs 20 integers using FOR…NEXT loop.
- calculates average of these integers.
- outputs the largest integer value and average of integers.

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Q.7. (Total 8 Marks)

EITHER

a.

i. Describe the purpose of each of the following components of computer: (3 Marks)

- Control unit
- Memory unit
- Arithmetic logic unit

ii. Sometimes, people prefer a hardcopy of a document instead of a softcopy. Give any FIVE reasons why this might be the case. (5 Marks)

OR

b. Discuss how computers play an effective role in accounting, communication, marketing and storage for smooth running of businesses. (8 Marks)

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END OF PAPER

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